

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In convex pentagon $ABCDE$, segment AB is parallel to segment DE . The measure of angle B is 139 degrees, and the measure of angle D is 174 degrees. What is the measure, in degrees, of angle C ?

Correct Answer: 47

Rationale

The correct answer is 47 . It's given that the measure of angle B is 139 degrees. Therefore, the exterior angle formed by extending segment AB at point B has measure $180 - 139$, or 41 , degrees. It's given that segment AB is parallel to segment DE . Extending segment BC at point C and extending segment DE at point D until the two segments intersect results in a transversal that intersects two parallel line segments. One of these intersection points is point B , and let the other intersection point be point X . Since segment AB is parallel to segment DE , alternate interior angles are congruent. Angle CXD and the exterior angle formed by extending segment AB at point B are alternate interior angles. Therefore, the measure of angle CXD is 41 degrees. It's given that the measure of angle D in pentagon $ABCDE$ is 174 degrees. Therefore, angle CDX has measure $180 - 174$, or 6 , degrees. Since angle C in pentagon $ABCDE$ is an exterior angle of triangle CDX , it follows that the measure of angle C is the sum of the measures of angles CDX and CXD . Therefore, the measure, in degrees, of angle C is $6 + 41$, or 47 .

Alternate approach: A line can be created that's perpendicular to segments AB and DE and passes through point C . Extending segments AB and DE at points B and D , respectively, until they intersect this line yields two right triangles. Let these intersection points be point X and point Y , and the two right triangles be triangle BXC and triangle DYC . It's given that the measure of angle B is 139 degrees. Therefore, angle CBX has measure $180 - 139$, or 41 , degrees. Since the measure of angle CBX is 41 degrees and the measure of angle BXC is 90 degrees, it follows that the measure of angle XCB is $180 - 90 - 41$, or 49 , degrees. It's given that the measure of angle D is 174 degrees. Therefore, angle YDC has measure $180 - 174$, or 6 , degrees. Since the measure of angle YDC is 6 degrees and the measure of angle CYD is 90 degrees, it follows that the measure of angle DCY is $180 - 90 - 6$, or 84 , degrees. Since angles XCB , DCY , and angle C in pentagon $ABCDE$ form segment XY , it follows that the sum of the measures of those angles is 180 degrees. Therefore, the measure, in degrees, of angle C is $180 - 49 - 84$, or 47 .